

Executive Summary – Social Inequality

This executive summary provides an overview of the project “Social Inequality – Trends over Time and Associations with Societal Factors”. The objective is to examine how social inequality evolves over time (i), how it differs across countries (ii), and how it is related to broader societal outcomes such as democracy, public safety, and healthcare access (iii). The analysis adopts a comparative international perspective and emphasizes both temporal dynamics and cross-country differences.

Data

The analysis is based on longitudinal country–year panel data from *Our World in Data*. The datasets cover income inequality (Gini index before and after tax), gender inequality (GII), democracy index (EIU- Democracy index), homicide rates, healthcare access (UHC-Service Coverage Index), and population size. To ensure comparability across indicators, all datasets were restricted to the common time window **2000–2023**. The resulting panel is unbalanced due to heterogeneous data availability across countries and indicators.

Problem and Methodology

Due to the large dataset, detailed analysis in some sections focuses on six countries, each located on a different continent, to illustrate cross-country differences and temporal trends in these indicators. Because it is difficult to compare indicators such as the Gini Index or the Gender Inequality Index on a global scale, their distributions were first analyzed using all available data. To account for country size, the Gini Index was weighted by each country’s population when estimating its density. This made it possible to identify typical values as well as extreme cases. To study the evolution of social inequality over time, line plots were used with years on the x-axis and indicator values on the y-axis. Boxplots were then applied to summarize the distributions over time, where the median indicated central tendency and the height of the box reflected variability. Finally, to examine the relationship between social inequality (Gini Index) and other variables, scatterplots based on country-level averages were employed. This ensured independent observations and allowed Spearman correlation coefficients to be computed, providing a quantitative assessment of associations.

Key Findings

The distribution plots indicate that most countries cluster around a Gini coefficient (After Tax) of approximately 0.41, reflecting moderate global income inequality. The Gender Inequality Index exhibits a bimodal distribution, with one peak at 0.21 and another at 0.58, highlighting the contrast between countries with low and high gender disparities. Scatterplots using country-level averages reveal the following relationships between the Gini Index and societal outcomes: a weak negative correlation with the Democracy Index (-0.261), indicating that higher inequality tends to coincide with weaker democratic performance; a strong positive correlation with the homicide rate (0.618), suggesting higher inequality is associated with higher crime levels; and a moderate negative correlation with healthcare access (-0.388), showing that greater inequality generally corresponds to reduced access to health services.

Outlook

Future research could expand the analysis by incorporating a broader set of countries and additional societal indicators. Addressing potential data gaps and inaccuracies, for instance due to political factors or missing information, would enhance the reliability of the results. Increasing the frequency and precision of measurements would allow for a clearer understanding of temporal trends. Despite these limitations, the findings provide a solid foundation for further examining the global impact of social inequality on democracy, crime, and healthcare access. Moreover, the analysis offers insights into how social inequality evolves over time and varies across countries.

Use of AI Tools

ChatGPT (OpenAI) was used to support linguistic refinement, code documentation, and technical problem-solving during the project. All suggestions were critically reviewed and adapted by the group. Responsibility for the analysis, interpretation, and final content remains entirely with the project team.